AI in Industry, weekend 2



Searching through representations

A story from ancient China

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



The court of King Xuan

The state of Qi, fourth century BC

King Xuan gathers scholars at his court. Good scholars are rare. Chunyu Kun, a witty advisor, brings him new ones.

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



The court of King Xuan

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How long would it take to find seven good scholars?

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



淳于髡一日而見七人於宣王。

In one day, Chunyu Kun presented seven men to King Xuan.

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



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King Xuan of Qi

reigned 319 to 301 BC

"Come. I have heard that one scholar in a thousand li is as if they stood shoulder to shoulder, and one sage in a hundred generations as if they came heel after heel. Now you present seven scholars in one morning: are scholars not plentiful, then?"

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Chunyu Kun

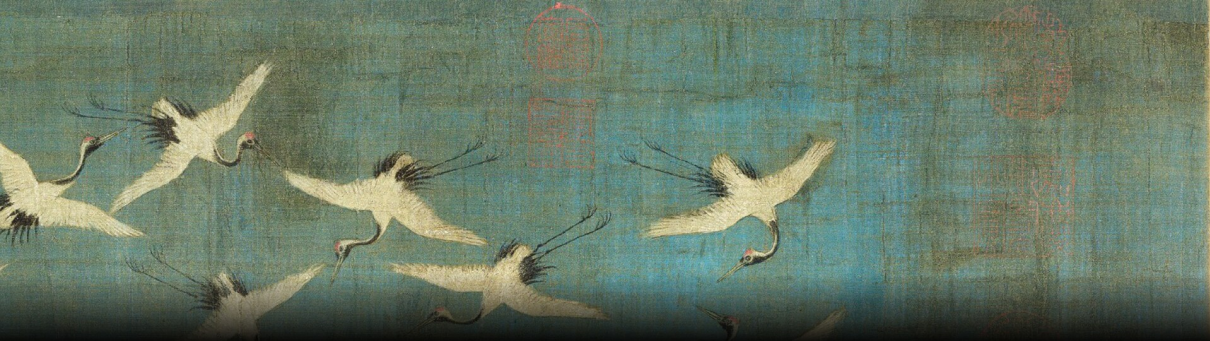
c. 386 to 310 BC

"Not so. Birds with the same wings live together; beasts with the same feet walk together. Look for chaihu and jiegeng in the marshes, and in generations you will not find one; go to the north slopes of the mountains 罩黍 and 梁父, and you load them by the cartful."

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer



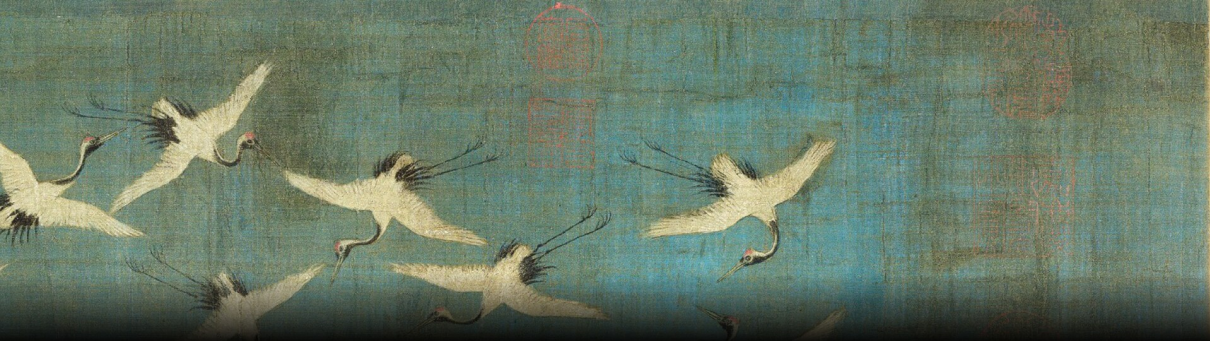
Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
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"Things each have their kind, and I, Kun, am of the kind of the worthy. Your Majesty seeking scholars from me is like drawing water from a river or taking fire from a fire-drill. I will present more; why only seven scholars?"

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Kun is himself a scholar, so scholars surround him.
To find more of a kind, look next to one you already know.

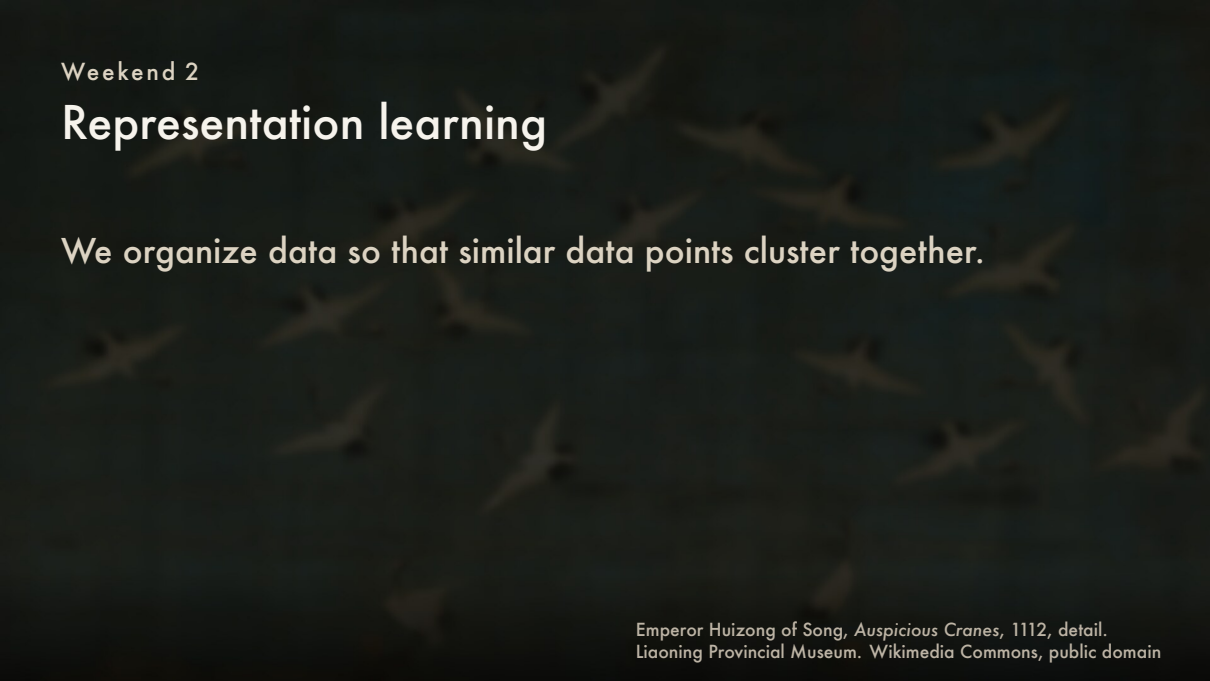
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Weekend 2

Representation learning

We organize data so that similar data points cluster together.



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Liaoning Provincial Museum. Wikimedia Commons, public domain

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For each data point, we compute a representation, its embedding: a list of numbers that captures its most characteristic features.

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Data points that belong together get similar embeddings.

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Liaoning Provincial Museum. Wikimedia Commons, public domain

Weekend 2

Representation learning today

Each side gets an embedding, and the most similar are matched.

Etsy a shopper's search with items for sale

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

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Etsy	a shopper's search with items for sale
Airbnb	a guest's search with homes

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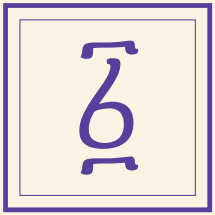
Representation learning is one of the main drivers of AI today.

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

Business cases

The inner product and nearest neighbors

Carlos Cotrini
AI in Industry, ETH Zurich
Friday 13 November 2026



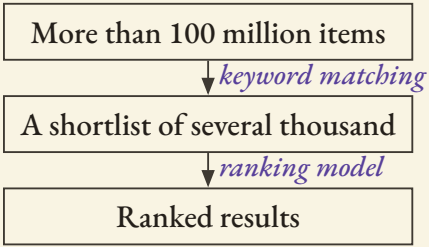
Section 1

Etsy

Etsy: a shortlist out of 100 million items

Etsy is a marketplace for handmade and vintage items.

Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.

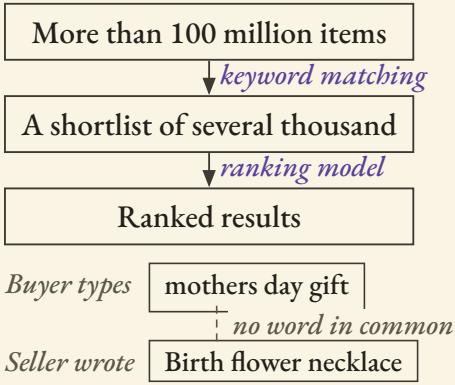


Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.

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The training data: a month of search logs

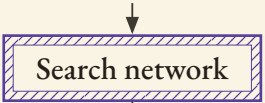
1 month: more than 150 million rows, from up to 30 million users.

Search	Who searched	Item	Action
mothers day gift	Zurich, searched necklaces	Flower necklace	purchase
bridge photos	New York, clicked print shops	Bridge print	cart add
wool socks	Oslo, searched knitting	Knitted socks	cart add

The rows are invented. Jha and others (Etsy), IEEE FMLDS 2024.

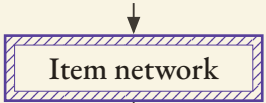
Every search and every item becomes a list of numbers

The search words and who searched



mothers day gift, in Zurich

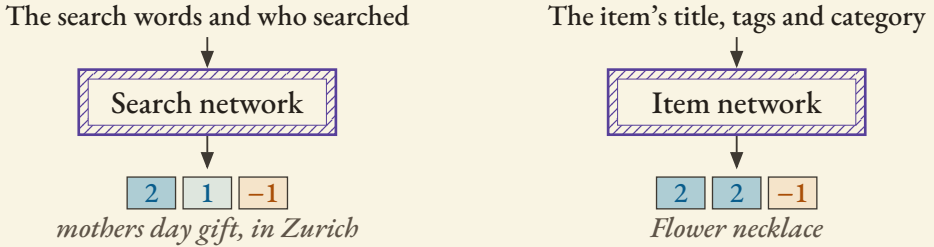
The item's title, tags and category



Flower necklace

Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

Every search and every item becomes a list of numbers



Training makes the inner product high for items bought or added to the cart after the search: here 7.

Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

A search arrives: sort by inner product

mothers day gift 2 1 −1

Invented numbers.

A search arrives: sort by inner product

mothers day gift	2	1	−1	*inner product*
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

Sorted: the top two join the shortlist

Flower necklace	7
Recipe board	4
Photo mug	3
Gift wrap	−4

Cost and value

Cost

72 hours *on 4 GPUs*

1 month *of search logs*

Value: A/B tests, relative change

+2.63% *visits that end in a purchase*

+5.58% *searches that end in a purchase*

Jha and others (Etsy), IEEE FMLDS 2024.

Cost and value

Cost

72 hours *on 4 GPUs*

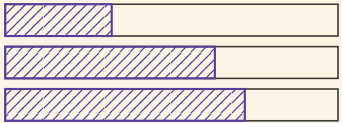
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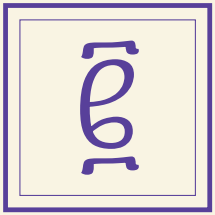
+2.63% *visits that end in a purchase*

+5.58% *searches that end in a purchase*

Popular searches: share of the bought products found in the top 100



Jha and others (Etsy), IEEE FMLDS 2024.

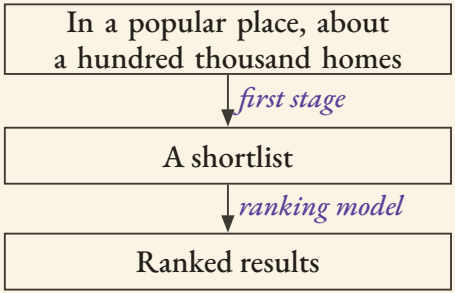


Section 2

Airbnb

Airbnb: the first stage of search

The first stage evaluates tens of thousands of homes.

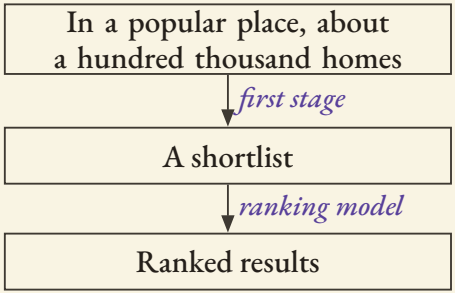


Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Airbnb: the first stage of search

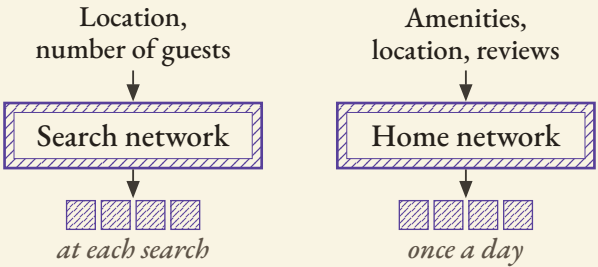
The first stage evaluates tens of thousands of homes.

It used to be only a simple linear model that scored every home once a day, whatever the search.



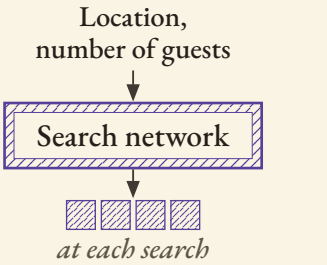
Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

Two networks and the nearest homes

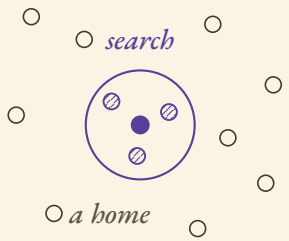


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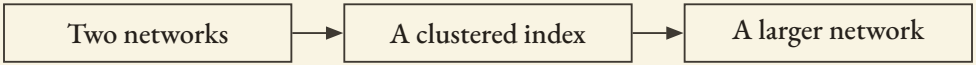


The nearest homes join the shortlist



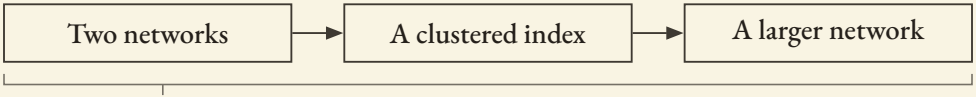
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The value: three A/B tests



Abdool and others (Airbnb), arXiv preprint 2026; Airbnb's blog 2025.

The value: three A/B tests



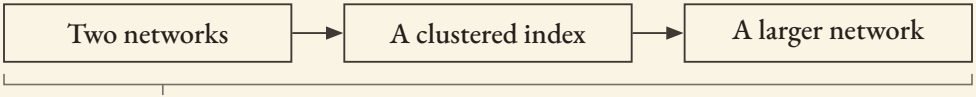
+0.31%

visits that end in a booking, all three, relative change

Airbnb calls this gain “a large portion of the team’s annual goal”.

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.

The value: three A/B tests



+0.31%

visits that end in a booking, all three, relative change

Airbnb calls this gain “a large portion of the team’s annual goal”.

+2.3% *bookings from promotional emails*

−16% *compute with the clustered index*

Abdool and others (Airbnb), arXiv preprint 2026; Airbnb’s blog 2025.